

Shiv Nadar University Chennai
CS3807 - Deep Learning Laboratory
Experiment 4

Comparative Study of Deep Convolutional Neural Network Architectures Using Transfer Learning

Degree & Branch	B.Tech Artificial Intelligence & Data Science	Semester	V
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1 Objective

To study the evolution of major CNN architectures, apply transfer learning and fine-tuning on CIFAR-10, and compare classification performance using the required evaluation measures.

2 Background Theory

Deep convolutional neural networks learn hierarchical spatial features directly from images. The architectures in this experiment mark important stages in the development of modern image-classification systems.

Model	Depth	Canonical Parameters	Main Contribution
LeNet-5	7	60K	First practical convolutional neural network
AlexNet	8	61M	ReLU, Dropout, data augmentation and GPU training
VGG16	16	138M	Deep network built from repeated 3×3 convolutions
GoogleNet	22	6.8M	Inception modules for parallel multi-scale feature extraction
ResNet50	50	25.6M	Residual learning with skip connections

LeNet-5. A shallow CNN developed for handwritten-digit recognition. Its small parameter count makes it fast and suitable for simpler image tasks.

AlexNet. AlexNet made deep CNNs practical at ImageNet scale by combining ReLU activations, Dropout, data augmentation and GPU training.

VGG16. VGG16 uses stacks of small 3×3 kernels. Repeated small convolutions increase depth and effective receptive field while keeping the architecture regular.

GoogleNet. Inception modules process several receptive-field sizes in parallel. 1×1 convolutions reduce channel dimensions before larger kernels, lowering computation while retaining multi-scale information.

ResNet. Residual blocks learn a residual mapping rather than the complete transformation:

$$F(x) = H(x) - x, \quad H(x) = F(x) + x.$$

The identity connection improves gradient flow and makes very deep networks easier to optimize.

Dilated Convolution. Kernel elements are spaced apart to enlarge the receptive field without increasing the number of kernel parameters.

Transpose Convolution. A learnable upsampling operation that increases spatial dimensions. It is commonly used in decoders, segmentation models and image-generation networks.

Transfer Learning. Features learned from a large source dataset such as ImageNet are reused for a new task. The pretrained convolutional base is first frozen while a new classifier is trained, then selected final layers are unfrozen for fine-tuning.

3 Dataset and Preprocessing

CIFAR-10 contains 60,000 RGB images in ten classes. The original 50,000-image training set was split into 45,000 training images and 5,000 validation images, while the standard 10,000-image test set was retained for final evaluation.

Item	Value
Training images	45,000
Validation images	5,000
Testing images	10,000
Image size	$32 \times 32 \times 3$
Classes	10

Pixel values were processed with the corresponding pretrained-model preprocessing function. The ten classes were Airplane, Automobile, Bird, Cat, Deer, Dog, Frog, Horse, Ship and Truck.

4 Experimental Procedure

1. CIFAR-10 was loaded, the dataset dimensions were verified, and one sample from each class was displayed.
2. An ImageNet-pretrained MobileNetV2 model was loaded without its original classifier. The convolutional base was frozen and followed by global average pooling, a 128-unit ReLU Dense layer and a 10-class Softmax layer.
3. The frozen-base classifier was trained for 10 epochs using Adam, learning rate 0.001, batch size 32 and categorical cross-entropy.
4. The final 20 MobileNetV2 layers were enabled for fine-tuning while Batch Normalization layers remained frozen. Training continued for 5 epochs at learning rate 10^{-5} .
5. Accuracy, precision, recall, F1-score, confusion matrix and the classification report were evaluated on the 10,000-image test set.
6. The listed hyperparameters were compared, and LeNet-5, AlexNet, VGG16, GoogleNet and ResNet50 were benchmarked under controlled CIFAR-10 settings.

5 Source Code

Repository: <https://github.com/Joshua-Divide/LAB>

6 Hyperparameter Study

The values required by the laboratory manual were evaluated. One factor was varied at a time from the frozen-base reference setting; partial freezing corresponds to the fine-tuning stage.

Setting	LR	Batch	Epochs	Opt.	Units	Frozen	Val. Acc.	Test Acc.
Baseline	0.001	32	10	Adam	128	All	30.52%	31.12%
Learning rate	0.0001	32	10	Adam	128	All	28.70%	29.32%
Batch size 16	0.001	16	10	Adam	128	All	30.56%	31.24%
Batch size 64	0.001	64	10	Adam	128	All	29.92%	31.03%
Epochs 20	0.001	32	20	Adam	128	All	31.20%	31.12%
Optimizer SGD	0.001	32	10	SGD	128	All	27.96%	28.12%
Dense units 256	0.001	32	10	Adam	256	All	30.50%	31.32%
Partial fine-tuning	10^{-5}	32	5	Adam	128	Partial	51.74%	52.07%

Partial fine-tuning produced the best test accuracy at 52.07%. Among frozen-base settings, 256 Dense units gave the highest test accuracy at 31.32%.

7 Mandatory Plots

7.1 Sample CIFAR-10 Images

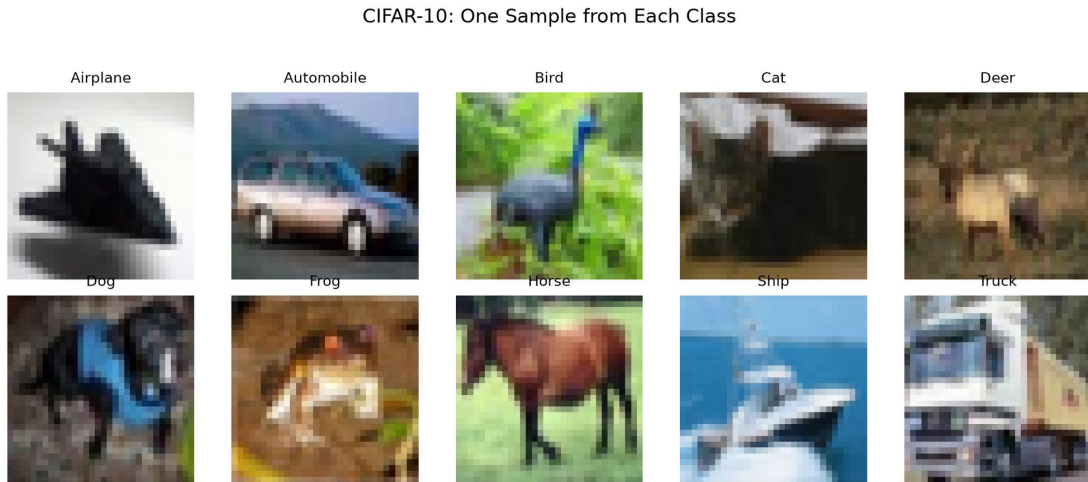


Figure 1: One sample image from each CIFAR-10 class.

Inference: The ten classes are visible, although several animal classes share similar low-resolution textures.

7.2 Training Accuracy

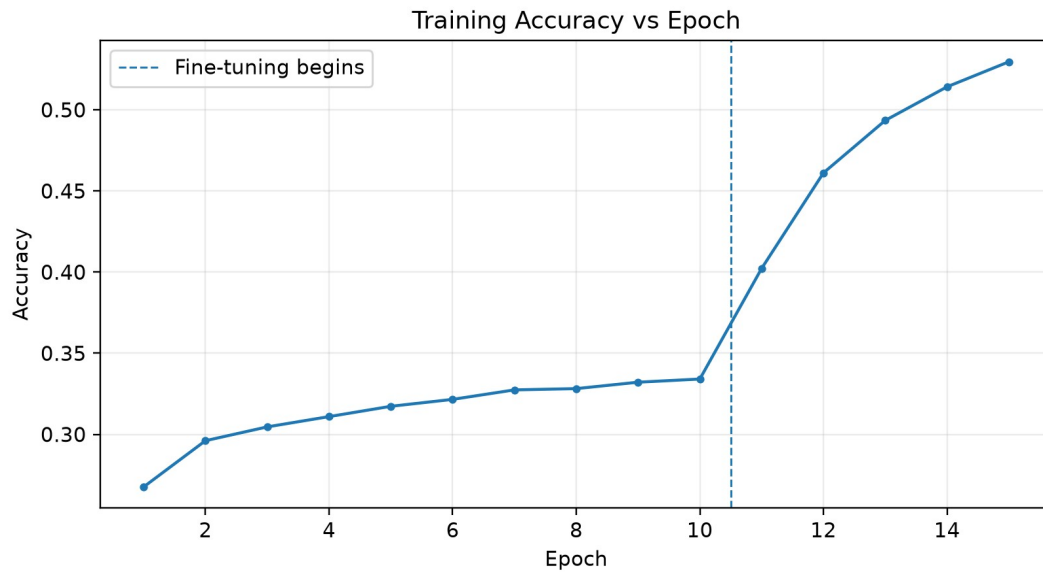


Figure 2: Training accuracy across the frozen and fine-tuning stages.

Inference: Training accuracy rises sharply after fine-tuning begins.

7.3 Validation Accuracy

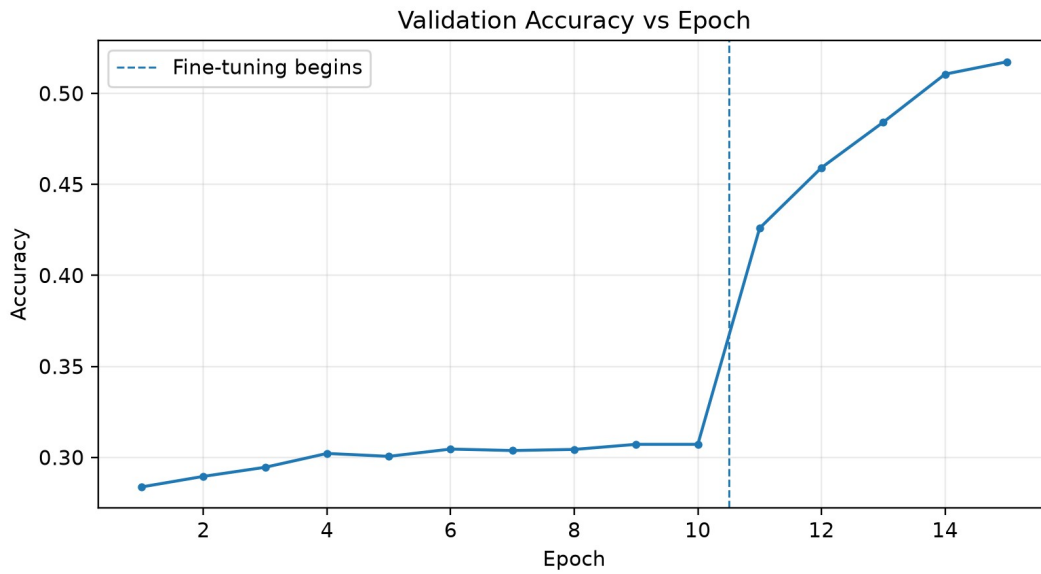


Figure 3: Validation accuracy across the frozen and fine-tuning stages.

Inference: Validation accuracy improves from about 30% to above 50% after fine-tuning.

7.4 Training Loss

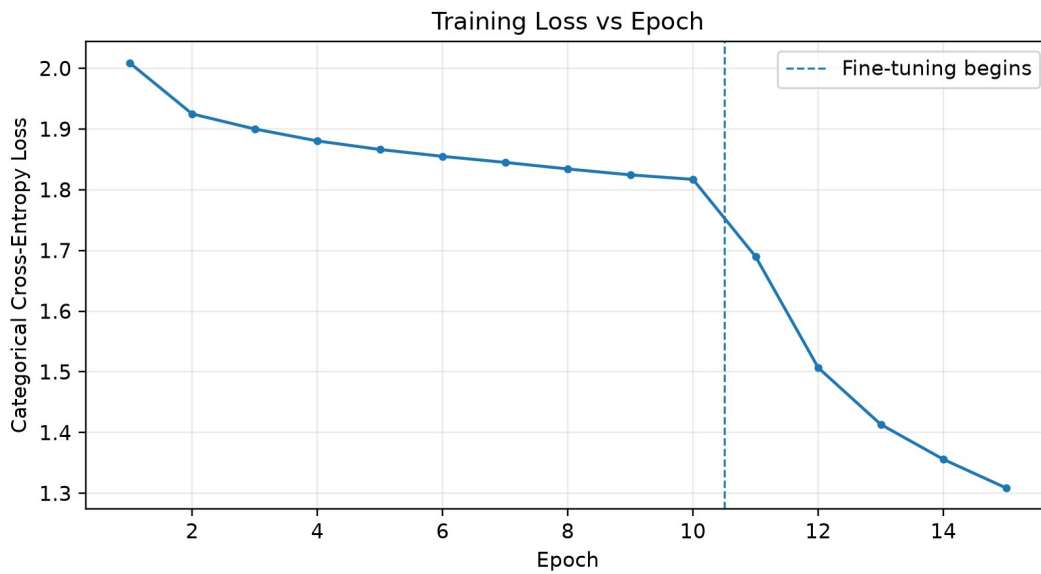


Figure 4: Training loss across the frozen and fine-tuning stages.

Inference: Training loss decreases throughout training, with a larger drop after unfreezing the final layers.

7.5 Validation Loss

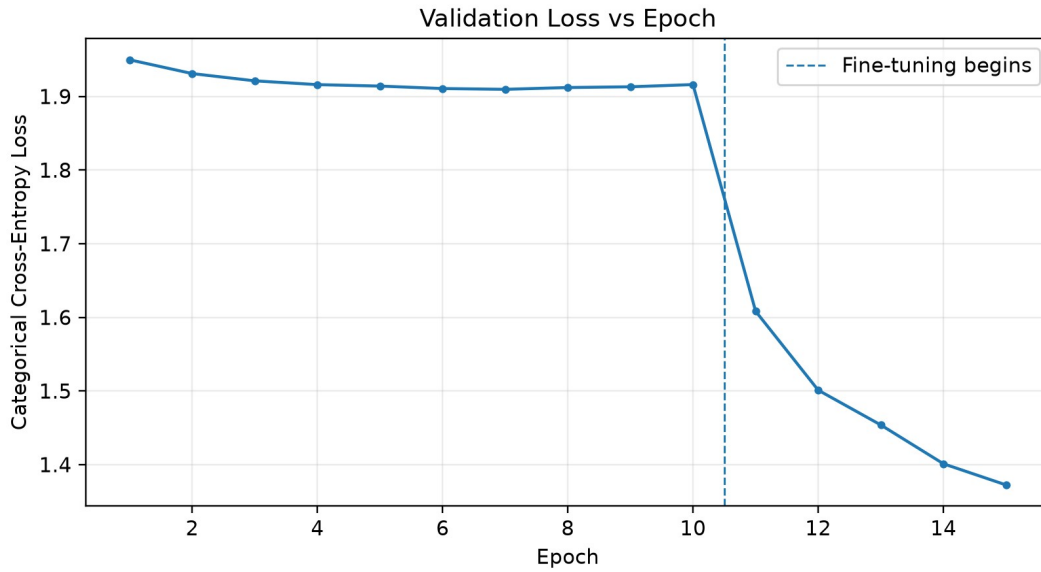


Figure 5: Validation loss across the frozen and fine-tuning stages.

Inference: Validation loss falls after fine-tuning, showing better performance on held-out data.

7.6 Confusion Matrix

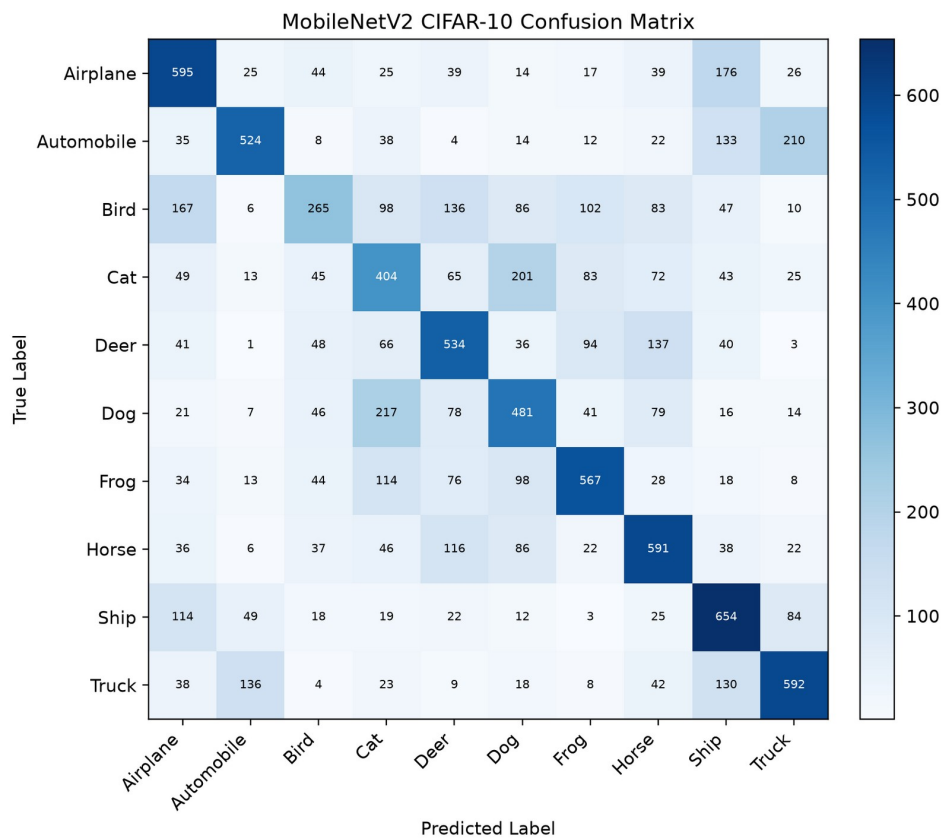


Figure 6: Confusion matrix of the fine-tuned MobileNetV2 model.

Inference: Most correct predictions lie on the diagonal; visually similar animal classes cause most of the remaining confusion.

8 Results

8.1 Performance Metrics

Metric	Value
Training Accuracy	52.96%
Testing Accuracy	52.07%
Precision (weighted)	0.5239
Recall (weighted)	0.5207
F1-score (weighted)	0.5169
Training Time	282.48 s
Total Parameters	2,423,242

Fine-tuning increased test accuracy from 30.46% to 52.07%, an improvement of 21.61 percentage points.

8.2 Comparison of CNN Architectures

The parameter counts below are for the implemented CIFAR-10 configurations rather than the canonical ImageNet models shown in the theory section.

Model	Implemented Parameters	Accuracy	Training Time
LeNet-5	62,006	38.15%	7.72 s
AlexNet	7,506,762	47.80%	256.83 s
VGG16	14,781,642	55.10%	262.17 s
GoogleNet	1,868,170	44.75%	391.83 s
ResNet50	23,851,274	61.25%	115.61 s

ResNet50 produced the highest measured accuracy in this controlled comparison.

8.3 Classification Report

Class	Precision	Recall	F1-score	Support
Airplane	0.5265	0.5950	0.5587	1000
Automobile	0.6718	0.5240	0.5888	1000
Bird	0.4741	0.2650	0.3400	1000
Cat	0.3848	0.4040	0.3941	1000
Deer	0.4949	0.5340	0.5137	1000
Dog	0.4598	0.4810	0.4702	1000
Frog	0.5975	0.5670	0.5818	1000
Horse	0.5286	0.5910	0.5581	1000
Ship	0.5050	0.6540	0.5699	1000
Truck	0.5956	0.5920	0.5938	1000
Weighted average	0.5239	0.5207	0.5169	10000

9 Additional Exercise Results

VGG16 and ResNet50 transfer learning

Model	Frozen Accuracy	Fine-Tuned Accuracy	Training Time
VGG16	52.85%	55.10%	262.17 s
ResNet50	59.95%	61.25%	115.61 s

Adam and SGD comparison

Optimizer	Validation Accuracy	Test Accuracy	Training Time
Adam	30.52%	31.12%	28.05 s
SGD	27.96%	28.12%	23.60 s

Frozen and fine-tuned comparison

Training Strategy	Test Accuracy	Training Time
Frozen convolutional base	30.46%	30.47 s
Partial fine-tuning	52.07%	252.01 s

Precision, recall and F1-score were evaluated for the fine-tuned model. The LeNet-5, AlexNet and ResNet50 comparison is included in the architecture benchmark above.

10 Discussion

Q	Answer
1	AlexNet was a breakthrough because it combined a deeper CNN with ReLU, Dropout, data augmentation and GPU training at ImageNet scale.
2	Stacked 3×3 filters give a large effective receptive field while adding more nonlinear transformations than one large filter.
3	Inception modules capture multiple spatial scales in parallel, while 1×1 convolutions reduce the computational cost.
4	Residual learning provides identity shortcuts that improve gradient flow and make very deep networks easier to optimize.
5	LeNet is shallow and lightweight for simpler tasks; ResNet is much deeper and uses residual blocks to train effectively.
6	Transfer learning reuses features learned on a large source dataset and adapts them to a new target task.
7	Fine-tuning adjusts selected pretrained layers so higher-level features better match the target dataset.
8	Dilated convolution enlarges the receptive field without adding kernel parameters; transpose convolution performs learnable spatial upsampling.
9	Pretrained models converge faster because useful low- and mid-level visual features are already available.
10	LeNet requires much less memory and computation. ResNet is more expensive, but residual connections allow much deeper feature learning.

11 Conclusion

Transfer learning and partial fine-tuning improved MobileNetV2 test accuracy from 30.46% to 52.07%. The required hyperparameter study, evaluation metrics, mandatory plots and additional architecture comparisons were completed. In the controlled CNN comparison, ResNet50 achieved the highest measured accuracy at 61.25%.

12 References

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